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Enhancement of methane production from anaerobic digestion of *Erigeron canadensis* via O_2-nanobubble water supplementation

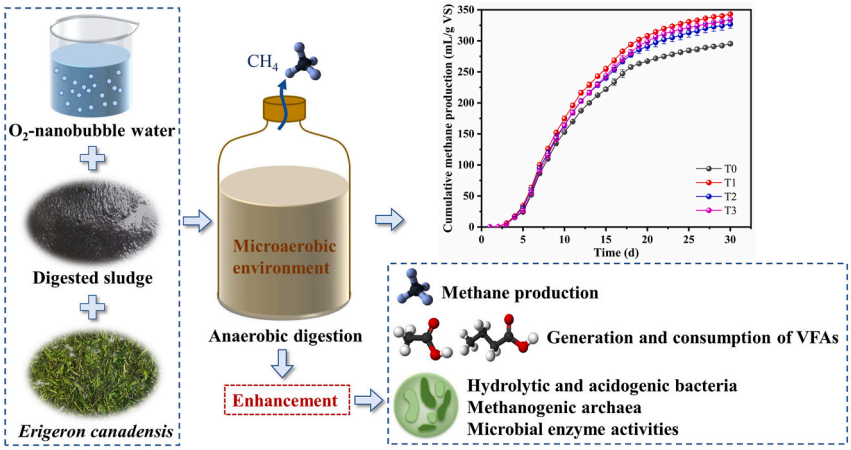
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HIGHLIGHTS

- *Erigeron canadensis* was firstly employed for methanogenesis in anaerobic digestion.
- Methane production was increased by 10.70–16.17% via O_2-NBW supplementation.
- The activity of microbial enzymes in O_2-NBW digesters was significantly enhanced.
- An enrichment of bacteria and archaea was presented via O_2-NBW addition.
- Addition of O_2-NBW could accelerate bacterial and archaeal metabolic activities.

GRAPHICAL ABSTRACT



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ABSTRACT

Malignant invasive *Erigeron canadensis*, as a typical lignocellulosic biomass, is a formidable challenge for sustainable and efficient resource utilization, however nanobubble water (NBW) coupled with anaerobic digestion furnishes a prospective strategy with superior environmental and economic effectiveness. In this study, influence mechanism of various O_2-NBW addition times on methanogenic performance of *E. canadensis* during anaerobic digestion were performed to achieve the optimal pollution-free energy conversion. Results showed that supplementation of O_2-NBW in digestion system could significantly enhance the methane production by 10.70–16.17%, while the maximum cumulative methane production reached 343.18 mL g^{-1} VS in the case of one-time O_2-NBW addition on day 0. Furthermore, addition of O_2-NBW was conducive to an increase of 2–90% in the activities of dehydrogenase, α-glucosidase and coenzyme F_{420}. Simultaneously, both facultative bacteria and methanogenic archaea were enriched as well, further indicating that O_2-NBW might be responsible for facilitating hydrolytic acidification and methanogenesis. Based on Kyoto Encyclopedia of Genes and Genomes (KEGG) cluster analysis, provision of O_2-NBW enhanced the metabolism of carbohydrate and amino acid, translation as

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well as membrane transport of bacteria and archaea. This study might offer the theoretical guidance and novel insights for efficient recovery of energy from lignocellulosic biomass on account of O_2-NBW adhibition in anaerobic digestion system, progressing tenor of carbon-neutral vision.

1. Introduction

Vigorous development of renewable energy sources is a paramount strategy to achieve energy security and transition in diverse countries throughout the world, while making contributions to alleviating the tension on fossil fuels as well as coping with the challenges of global climate change (Bian et al., 2021). One of the foremost renewable energy sources, in a long-term perspective, is biomass resource, because of its substantial richness, variety of transformations and great significance to carbon neutrality (World Bioenergy Association, 2021). Driven by this circumstance, invasive alien plants such as horseweed (*Erigeron canadensis*), water hyacinth (*Eichhornia crassipes*), alligator weed (*Alternanthera philoxeroides*) and cordgrass (*Spartina alterniflora*), as promising biomass resources due to their low cost and tremendous reserves, offer excellent occasions for the sustainable conversion of energy (Li et al., 2020).

Erigeron canadensis, popularly known as horseweed, is a fleetly colonizing invasive annual weed worldwide, especially in China and Europe where it has been named as a critical weed for management and control (Galkina and Vinogradova, 2020). Originally from North America, however, this notorious weed spreads at a swifter speed via more prolific achenes (over 200,000 per plant) on account of stronger fecundity (Bhattacharya et al., 2022), in addition to be characterized by a briefer growth cycle and higher niche breadth. Moreover, the invasive *E. canadensis* can supersede native biota to occupy a dominant position within terrestrial ecosystems, including cultivated lands, pastures, hillsides, wildernesses, and river valleys. Detrimentally, invasion of *E. canadensis* is accompanied by the breeding of pests as a host, the loss of biodiversity due to secretion of allelochemicals (e.g., polyphenols, terpenoids and flavonoids), and the significant decline in the yield of staple crops (e.g., corn, cotton, soybean, sugarbeet, etc.) of around 28–83% (Weaver, 2001). At present, the control strategies involved in *E. canadensis* are either unsustainable chemical measures like herbicides application, not economical physical measures like manual digging, or prolonged biological measures like obligate predators import, however, the net acreages encroached by its invasions continues to enlarge (Szabo et al., 2020). Hence, there is an urgent demand for sustainable integrated management and efficient resource utilization of *E. canadensis*.

With the advantage of being rich in cellulose but low in lignin contents, *E. canadensis* is a powerful attractive candidate as a feedstock for anaerobic digestion. Furthermore, the anaerobic digestion treatment can not only accomplish low-cost and sustainable management of *E. canadensis*, but also attain the purpose of efficiently converting it into methane, an alternative renewable energy, which possesses superior environmental and economic effectiveness (Saha et al., 2020). Although the fast-growing *E. canadensis* is an ideal accessible lignocellulosic biomass for anaerobic digestion, a primary drawback is that the hydrolysis stage is rate-limited on account of its recalcitrant cell walls (Song et al., 2020). To overcome this obstacle, disparate strategies focused on enhancing methane production from anaerobic digestion, for example, pretreatments prior to digestion, improvements in configuration of digesters, and dosing of diverse additives, have been attempted intensively (Sawatdeenarunat et al., 2015; Guan et al., 2022; Haque et al., 2022). Nonetheless, the supernal energy and technology requirements of above treatments, coupled with the potential for secondary pollution, are neither cost-effective nor eco-friendly. It is imperative to search for novel strategies to address clean transformation of bioenergy.

Lately, NBW has drawn more and more attention as an additive with unique preponderance to advance the performance of anaerobic digestion of lignocellulosic biomass. Due to the enrichment of nanobubbles (NBs) with a diameter of less than 1000 nm in NBW, it is peculiar to greater specific surface area, faster mass transfer efficiency, longer residence time, higher surface charge, stronger gas solubility, more reactive oxygen species (ROS) and so on (Liu et al., 2016; Michailidi et al., 2020; Xiao and Xu, 2020). In the case of the primary forms of NBs in NBW, they are either surface bubbles that construct a gas-filled space at the interface of solid substrates, or free bubbles that undergo Brownian motion in the liquid phase (Theodorakis and Che, 2019; Kim et al., 2020). Up to the present, with respect to a variety of NBs types in NBW, NBs of O_2, Air, H_2, N_2, and CO_2 have been studied to achieve high-efficiency promotion of methane production (Chuenchart et al., 2021). In the research by He et al. (2022), methane production, hemicellulose and cellulose degradation rates of corn straw were increased by 12–25%, 13–26% and 20–33% in digesters supplemented with NBW, respectively. In general, excessive introduction of O_2 is deemed to have great negative impacts on anaerobic microorganisms. However, the addition of NBW containing O_2 can achieve a delicate balance without generating over aeration or exceeding the antioxidant capacity of microorganisms (Wang et al., 2021). Strikingly, both Air-NBW and O_2-NBW can create a microaerobic environment to tremendously accelerate the hydrolytic acidification stage of substrates, thereby boosting methane production (Wang et al., 2020b; Hou et al., 2021). Compared with conventional microaeration technologies, O_2-NBW is capable of persistently maintaining a microaerobic environment via highly stable O_2-NBs (Ebina et al., 2013). The prominent superiority of microaerobic environment is to heighten the diversity and metabolic activities of facultative bacteria that are involved in hydrolysis, and at the same time expedite the consumption of intermediates to stabilize anaerobic digestion system (Zhu et al., 2009). In addition, O_2 freed from O_2-NBW zealously serves as an electron acceptor to motivate microbial activities by correspondingly accelerating electron transfer (Wang et al., 2020b). Nevertheless, a more in-depth study on the particular mechanism for O_2-NBW provision to methanogenesis of complicated lignocellulosic biomass in anaerobic digestion system has not yet been fulfilled, in terms of disparate metabolic pathways of microorganisms, correlative enzyme activities, and carbon balance of the system.

Therefore, this study investigated the influence of various O_2-NBW addition times on methane production from *E. canadensis* which was firstly reported as the feedstock for anaerobic digestion. The prime purposes of this work are: (i) to monitor diverse conventional indicators of digestion system to explore the impact of O_2-NBW addition on performance of anaerobic digestion; (ii) to characterize the function of O_2-NBW by measuring interrelated microbial enzyme activities; (iii) to expound underlying mechanisms based on exposure of bacterial and archaeal communities under O_2-NBW provision; (iv) to assess carbon flow distribution in anaerobic digestion system supplemented with O_2-NBW. To progress tenor of carbon-neutral vision, the results of this study are supposed to offer theoretical guidance and novel insights for efficient recovery of energy from lignocellulosic biomass (*E. canadensis*) on account of O_2-NBW adhibition in anaerobic digestion system.

2. Materials and methods

2.1. Feedstock and inoculum

E. canadensis was collected from abandoned farmland in Pengjiazhuang Village, Weifang, Shandong, China. Before further treatment, *E. canadensis* was dried, ground to pass through a 2 mm sieve and

stored in sealed plastic bag at 4 °C. Digested sludge was sampled from the Xianyang Road Wastewater Treatment Plant in Tianjin, China. The sludge was incubated at 35 °C for 2 weeks prior to employ to activate the microorganisms. At the beginning of incubation, the total solids (TS) and volatile solids (VS) contents of digested sludge were 4.84 wt% and 2.18 wt%, respectively. Characteristics of feedstock (*E. canadensis*) and inoculum (sludge) are presented in Table S1.

2.2. O_2-NBW

The O_2-NBW used in this study was prepared by a NB generator (YC-MNB–III–0.15, Yuchuang Environmental Technology Co., Ltd., China) adopting a gas-water circulation method. O_2 (purity ≥99.995%) was purchased from Tianjin Anxing Industrial Gas Sales Co., Ltd., China. Deionized water (DW) circulation flow rate, gas intake velocity and pressure of the NB generator were respectively controlled at 2.5 L min^{-1}, 0.07 L min^{-1} and 0.24 MPa for 20 min to obtain O_2-NBW. In this experiment, the bubble concentration and average particle size of O_2-NBs were 2.6 × 10^8 mL^{-1} and 189.4 nm, respectively.

2.3. Experimental design

4.5 g of *E. canadensis* and 200 mL of inoculum sludge (S/I = 4:3) were added to each 500 mL serum bottle, in which working volume was set as 400 mL. In this study, both DW and O_2-NBW total volumes were controlled at 150 mL for each digester. The different addition times were T0 (DW, on day 0, 150 mL each time, control group), T1 (O_2-NBW, on day 0, 150 mL each time), T2 (O_2-NBW, on day 0, on day 6, 75 mL each time), T3 (O_2-NBW, on day 0, on day 6, on day 12, 50 mL each time), respectively. DW was exploited when adding O_2-NBW on days 0, 6, and 12 to ensure a consistent volume of liquid in each digester. To eliminate the biogas generated by the inoculum sludge, a blank with the same amount of sludge was added. Initial pH was adjusted to 7.20 with 2 M NaOH or HCl while an anaerobic environment was created by purging with nitrogen for 5 min. The experiments were carried out in a constant temperature incubator at 35 °C for 30 days with manual shaking once a day. Each group of experiments was terminated until the daily methane production remained less than 1% of the total methane production for 5 consecutive days. All experiments were performed in triplicate.

2.4. Analytical methods

TS and VS were analyzed based on standard methods (Eaton et al., 2005). Elemental composition (C, H, N and S) of feedstock and inoculum was analyzed by an elemental analyzer (Elementar Vario ELIII, Germany). Lignin, cellulose and hemicellulose were determined by Van Soest method using a fiber analyzer (A2000I, ANKOM, USA). The pH of biogas slurry was measured by a pH meter (ST3100/F, Ohaus Starter, USA). The concentration of ammonia nitrogen was determined by Nessler's reagent colorimetric method (HJ 535–2009, China). Biogas production was measured using a 60 mL syringe and then converted to volume at standard temperature and pressure (STP). Biogas composition (CH_4 and CO_2) was determined by a gas chromatograph (GC) (8860, Agilent, USA) packed with a thermal conductivity detector (TCD) and an HQ + 5A column (50 °C), as well as the resulting chromatogram of which shown in Fig. S1. Injector and detector temperatures were set to 80 °C and 250 °C, respectively. Helium as carrier gas was adjusted to a flow rate of 5 mL min^{-1}. Liquid samples for analysis of volatile fatty acids (VFAs) were extracted via a 2.5 mL syringe, centrifuged for 5 min at 15,000 rpm and filtered through a 0.22 μm membrane. In order to acidify the filtrate, 20% (v/v) phosphoric acid (25% v/v) was added before being measured by a gas chromatograph (8860, Agilent, USA) equipped with DB-FATWAX UI column (30 m × 250 μm × 0.25 μm) and flame ionization detector (FID). Ingredients of VFAs in terms of acetic acid, propionic acid, isobutyric acid, butyric acid, isovaleric acid, and valeric acid were detected. The flow rate of N_2 carrier gas was maintained at 25 mL min^{-1}, while the temperatures of injector and detector were 250 °C and 300 °C, respectively. Two microbial enzymes (on day 6), including dehydrogenase (DHA) and α-glucosidase (α-GC), along with coenzyme F_{420} (on day 12), were assayed utilizing ELISA kit provided by Beijing ZKGX Research Institute of Chemical Technology, China.

2.5. Theoretical methane production and biodegradability

Buswell's equation was employed to estimate the theoretical methane production (TMP) of *E. canadensis*, as shown in Eqs. (1) and (2) (Wang et al., 2020b).

$$C_aH_bO_c + \left(\frac{4a\text{-}b\text{-}2c}{4}\right)H_2O \rightarrow \left(\frac{4a + b\text{-}2c}{8}\right)CH_4 + \left(\frac{4a\text{-}b + 2c}{8}\right)CO_2 \quad (1)$$

$$TMP = \frac{22400\ (a/2 + b/8\text{-}c/4)}{12a + b + 16c} \quad (2)$$

where *a*, *b* and *c* represent the molar fractions of C, H and O, respectively.

Experimental methane production (EMP) and TMP can be exploited to calculate the anaerobic biodegradability (BD) of samples according to Eq. (3) (Aragón-Briceño et al., 2017).

$$BD(\%) = \frac{EMP}{TMP} \times 100 \quad (3)$$

2.6. Model-based analysis

In order to further explore the dynamic analysis of cumulative methane production during anaerobic digestion, the modified Gompertz model as in Eq. (4) (Song et al., 2021a) was adopted for fitting.

$$M_{(t)} = P \times \exp\left\{-\exp\left[\frac{e \times R_{max}(\lambda\text{-}t)}{P} + 1\right]\right\} \quad (4)$$

where $M_{(t)}$ (mL g^{-1} VS) refers to the cumulative methane production at digestive time *t* (d). *P* (mL g^{-1} VS) represents the ultimate methane production. R_{max} (mL g^{-1} VS d^{-1}) denotes the maximum methane production rate. λ (d) stands for the lag phase of anaerobic digestion. *e* means Euler's number.

2.7. Microbial community analysis

Biogas slurry was collected for analysis of bacterial and archaeal communities after 30 days' anaerobic digestion. After extracting microbial DNA, the V3–V4 regions of bacteria were amplified with primer sequences 338F (5′-ACTCCTACGGGAGGCAGCAG-3′) and 806R (5′-GGACTACHVGGGTWTCTAAT-3′). Furthermore, the primer sequences 524F10extF (5′-TGYCAGCCGCCGCGGTAA-3′) and Arch958RmodR (5′-YCCGGCGTTGAVTCCAATT-3′) were used to perform PCR amplification on the V3–V4 regions of the archaea. The subsequent treatments of PCR products were performed by Shanghai Majorbio Bio-pharm Technology Co., Ltd., China, namely purification, quantification, establishment of sequencing library, and sequenced on the Illumina MiSeq platform (Illumina, San Diego, USA). Detailed steps were previously described (Song et al., 2021b).

2.8. Statistical analysis

In this work, all data were calculated employing Microsoft Office Excel 2019. The experimental data were expressed as mean ± standard deviation. SPSS 12.0 was used to conduct one-way analysis of variance (ANOVA) to analyze the significance of obtained results, and significant difference was assumed at $p < 0.05$. Origin 2021 was applied for all figures.

3. Results and discussion

3.1. Effect of O_2-NBW addition on methane production from *E. canadensis*

3.1.1. Daily and cumulative methane production

Changes of daily and cumulative methane production along with associative methane content of *E. canadensis* during anaerobic digestion under different O_2-NBW addition times are presented in Fig. 1. At least 90% of general methane production was achieved within 20 days of initiation of anaerobic digestion. Over the digestion period of 30 days, the daily methane production of digesters with O_2-NBW was higher than that of digesters with DW. As can be seen from Fig. 1(a), the maximum daily methane productions of T1, T2 and T3 with O_2-NBW addition were 36.79 ± 1.29 mL g^{-1} VS d^{-1}, 36.08 ± 0.74 mL g^{-1} VS d^{-1} and 35.34 ± 1.89 mL g^{-1} VS d^{-1} on day 7, respectively, while that of T0 with DW addition was 34.12 ± 1.09 mL g^{-1} VS d^{-1}. According to Fig. 1(c), the increase of methane content in the earlier stage was accompanied by the gradual increase of O_2-NBW addition such as T1 with a 16.58% improvement in methane content compared to T0 during day 4 to day 6. Noteworthy, due to the addition of fresh O_2-NBW on day 6, the methane content of T3 was higher than that of other digesters in the later stage, reaching as high as 69.46% on day 19. Thus, it is deduced that the presence of O_2-NBW can promote higher methane content and better biogas quality, which is beneficial for energy recovery.

As shown in Fig. 1(b), there was significant impact ($p < 0.05$) of O_2-NBW addition of digesters to enhance cumulative methane production compared with DW addition. In this regard, compared to T0, the cumulative methane production of T1, T2 and T3 groups were increased by 16.17%, 10.70% and 13.30%, respectively. Similarly, Wang et al. (2020b) showed that an increase (29.9%) in cumulative methane production from cellulose was obtained in the O_2-NBW reactors. In addition, the maximum cumulative methane production of 343.18 ± 1.66 mL g^{-1} VS was observed in T1 digester with one-time addition of O_2-NBW only on day 0. Presumable reasons can be explained: (i) bursting of O_2-NBs creates a microaerobic environment to strengthen the diversity and competence of facultative bacteria (as mentioned in 3.4), thereby promoting feedstocks hydrolysis (Song et al., 2023), (ii) existence of O_2-NBs with higher surface charge improves the efficiency of gas transfer as well as the activity of electron transport system (ETS) (Chuenchart et al., 2021). In general, the anaerobic digestion system coupled with O_2-NBW effectively achieved the maximum methane production of *E. canadensis*, however, the correlative mechanisms are necessary to further investigated.

3.1.2. Kinetic analysis of methane production

The fitting parameters based on the modified Gompertz model were listed in Table 1. The calculated P values were in good agreement with the experimental $M_{(t)}$ values, indicating that experimental methane productions were commendably fitted with the modified Gompertz model, with the supernal correlation coefficient (R^2) greater than 0.996. T1 exhibited a remarkable rise in both maximum methane production and methane production rate, with the 16.77% and 11.50% higher value than T0, respectively. Moreover, the R_{max} values were significantly different between the O_2-NBW-added and DW-added digesters ($p < 0.05$), which further demonstrated that the addition of O_2-NBW could be more conducive to methane production. However, O_2-NBW supplementation had no significant influence ($p > 0.05$) on the lag phase.

On the basis of elemental analysis data and equations, the theoretical molecular formula and corresponding TMP of *E. canadensis* were calculated as $C_{3.53}H_{5.33}O_{2.08}$ and 445.29 mL g^{-1} VS, respectively. After 30 days of anaerobic digestion, the BD values (indicator for evaluating the degradation of organic matters) followed the descending order as T1 > T3 > T2 > T0 (Table 1). The differences among the four groups might be attributed to the generation of ·OH from O_2-NBs in the O_2-NBW, which could be conducive to the breakage of C–C and C–O bonds within

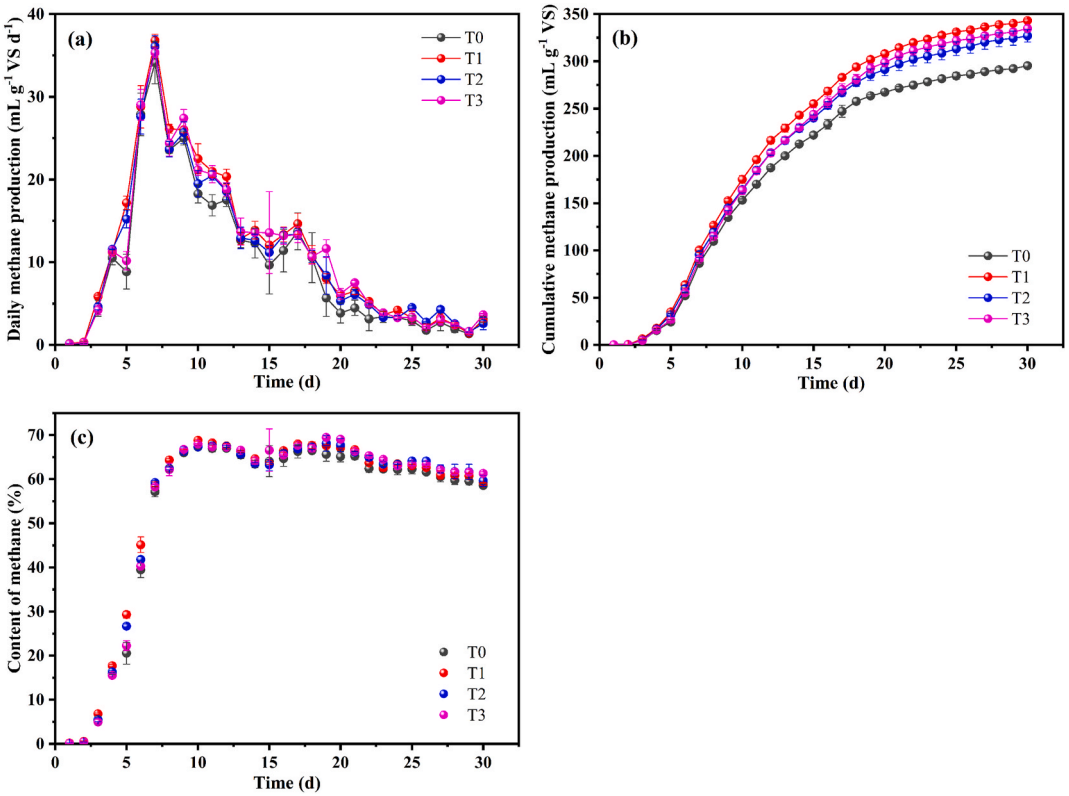


Fig. 1. Methane production from *E. canadensis* during anaerobic digestion under different O_2-NBW addition times. (a) Daily methane production, (b) Cumulative methane production and (c) Methane content.

Table 1
Experimental data and fitting parameters based on the modified Gompertz model.

Groups	TMP (mL g^{-1} VS)	EMP (mL g^{-1} VS)	BD (%)	Modified Gompertz model			
				P (mL g^{-1} VS)	R_{max} (mL g^{-1} VS d^{-1})	λ (d)	R^2
T0	445.29	295.41 ± 3.04	66.34	291.42 ± 2.42	23.31 ± 0.66	3.67 ± 0.18	0.9970
T1	445.29	343.18 ± 1.66	77.07	340.28 ± 2.82	25.99 ± 0.70	3.51 ± 0.17	0.9972
T2	445.29	327.03 ± 6.67	73.44	323.46 ± 2.97	24.33 ± 0.70	3.49 ± 0.19	0.9967
T3	445.29	334.70 ± 3.63	75.16	332.89 ± 2.94	24.83 ± 0.67	3.72 ± 0.18	0.9972

TMP: theoretical methane production; EMP: experimental methane production; BD: biodegradability; VS: volatile solids.
P: ultimate methane production; R_{max}: maximum methane production rate; λ: lag phase time; R^2: correlation coefficient.

lignin molecules, thereby advancing the anaerobic biodegradability of refractory lignocellulosic biomass (Yang et al., 2019). The overall consequences represented an increase in methane production from *E. canadensis* with the addition of O_2-NBW.

3.2. Variation of O_2-NBW addition on microbial enzyme activities

α-Glucosidase (α-GC) is responsible for crushing α-1,4-glucosidic linkage to hydrolyze oligosaccharides and disaccharides (Gao et al., 2020). Thus, α-GC, as a key enzyme straightforward determining the hydrolysis of primary organic fraction of *E. canadensis*, which is carbohydrate, was selected to characterize the effect of O_2-NBW on enzyme activity during anaerobic digestion. Seen from Fig. 2(a), the provision of O_2-NBW improved α-GC activity to a certain extent, ranging from 1.53% to 8.36%. The results suggested that O_2-NBW supplementation could enrich facultative bacteria (as noted in 3.4) and generate superior extracellular enzymes to expedite the hydrolysis, thereby offering abundant digestive substances for subsequent methanogens to indirectly boost methane production (Ruan et al., 2019; Li et al., 2021). As an intracellular enzyme, dehydrogenase (DHA) is served as a measurement of electron transfer activity and holistic microbial activity during anaerobic digestion on account of its ability to catalyze dehydrogenation reactions (Wang et al., 2017). Notably, after O_2-NBW addition, DHA activity in all digesters was significantly augmented ($p < 0.05$) compared to the control digester, confirming that the microbial activity was enhanced by O_2-NBW. Specifically, the maximum DHA activity of 220.84 ± 5.14 IU L^{-1} in T1 group is shown in Fig. 2(a), which was 38.73% higher than that in T0 group. Therefore, a new strategy for achieving higher enzyme activities of anaerobic digestion system is to afford O_2-NBW, thereby advancing methane production.

Coenzyme F_{420}, an electron transfer carrier peculiar to hydrogenotrophic methanogens, is generally applied for revealing methanogenic activity (Deng et al., 2021). The significant enhancement ($p < 0.05$) of coenzyme F_{420} activity during anaerobic digestion by the addition of O_2-NBW is distinctly shown in Fig. 2(b), which was consistent with the study by Fan et al. (2021b). On day 12, the activities of coenzyme F_{420} in T1, T2 and T3 were increased by 89.95%, 14.69%, and 65.41% in comparison to the control group (T0), respectively. This indicated that O_2-NBW supplementation could facilitate electron transfer in digestion system while enhancing the activity of hydrogenotrophic methanogens, resulting in elevated methane production. Overall, the stimulation of O_2-NBW on enzyme activities can accelerate not only hydrolysis but also methanogenesis, aiming to aggrandize the methane production from *E. canadensis*.

3.3. Removal of VS, volatile fatty acids and pH during anaerobic digestion

VS removal rate is supposed to display the hydrolysis and conversion efficiency of substrate in anaerobic digestion system. The changes of VS content and VS removal rate in various groups are shown in Fig. S2. Compared with T0 control group, the groups added with O_2-NBW emerged a higher VS removal rate, of which the T1, T2 and T3 groups were 39.20%, 38.75% and 39.11%, respectively. Advancement of VS removal rate signified that the motivate effect of O_2-NBW on anaerobic digestion of *E. canadensis* might be derived from the promotion of substrate hydrolysis as well as achieving energy conversion efficiently.

VFAs are intermediates generated after the hydrolytic and acidogenic stages, which play a crucial role in the subsequent methanogenesis phase. Components of VFAs in term of acetic acid, propionic acid, isobutyric acid, butyric acid, isovaleric acid, and valeric acid are exhibited in Fig. 3. VFAs were abundantly inspired in all digesters after initiation of anaerobic digestion process, peaked at 4–6 days, and then leveled off after 15 days. Dissimilarly, higher VFAs concentration was manifested in the digesters supplemented with O_2-NBW during the first 15 days, compared to the control group, which was statistically significant ($p < 0.05$). In particular, the O_2-NBW groups aggrandized 38.41–56.99% more than the DW group in VFAs on day 6. Due to the presence of a small amount of oxygen, an increase in extracellular hydrolases manufacture from a more luxuriant community of hydrolytic bacteria can enhance hydrolytic acidification to produce greater VFAs (Nguyen and Khanal, 2018). Likewise, a study by Fan et al. (2021a) showed that Air-NBW

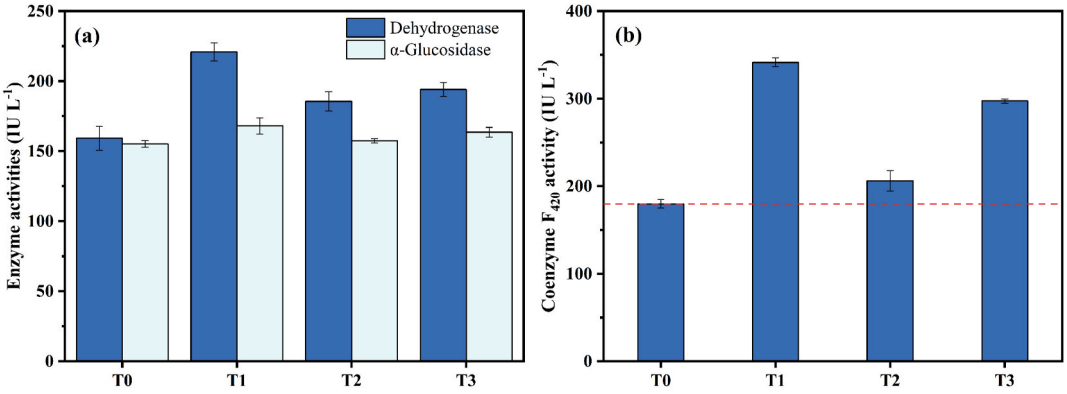


Fig. 2. Variation of O_2-NBW addition on microbial enzyme activities during anaerobic digestion. (a) Dehydrogenase (DHA) and α-Glucosidase (α-GC), (b) Coenzyme F_{420}.

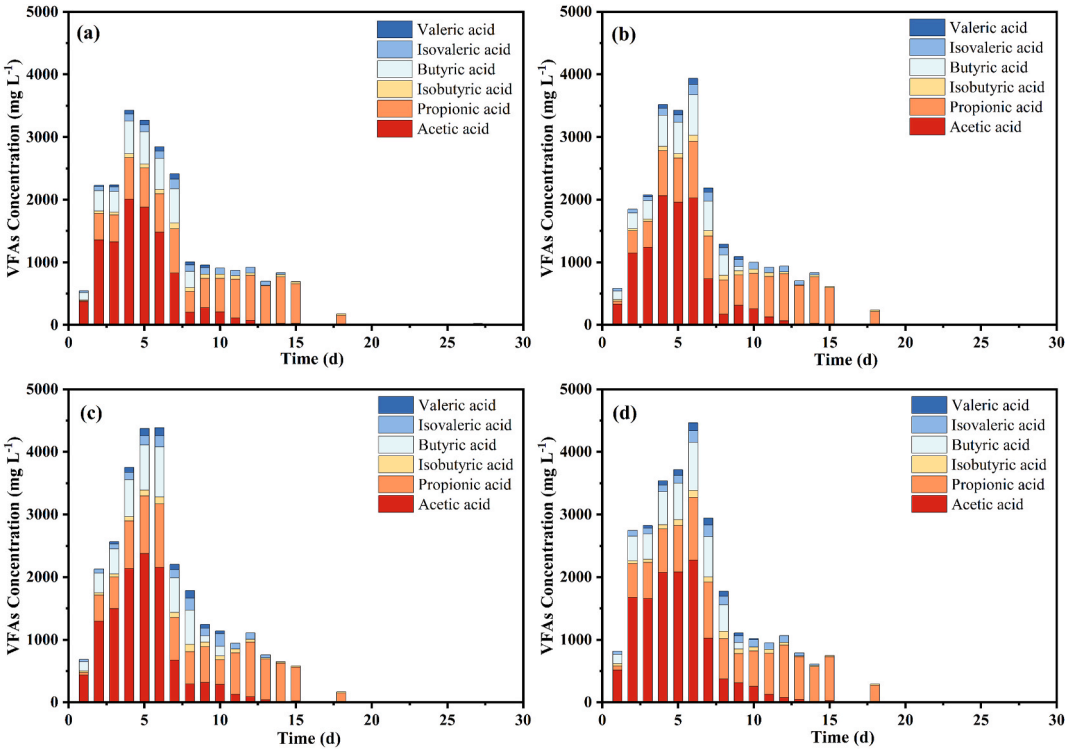


Fig. 3. VFAs components in each digester during 30-day anaerobic digestion. (a) T0, (b) T1, (c) T2, (d) T3.

supplementation accelerated VFAs yield during the hydrolysis-acidification phase. On the other hand, a speediness depletion of VFAs, presumably by methanogens, was observed without inhibition on days 7–9. Correspondingly, from day 7 to day 9, the methane production in T1, T2 and T3 groups were increased by 14.48%, 8.76% and 5.09% in contrast with T0, respectively. Hence, it is further deduced that the addition of O_2-NBW might lead to faster not only production but also consumption of VFAs in hydrolytic acidification process, which in turn assisted in maximizing the subsequent methane production.

Regardless of the digester, acetic acid, propionic acid and butyric acid were the predominant VFAs. Among them, acetic acid and butyric acid, which is a conversion source of acetic acid, positively contribute to methane production of methanogens (Komemoto et al., 2009). The concentrations of both acetic acid and butyric acid first increased and then decreased along with anaerobic digestion, and were lower than 80 mg L^{-1} after 12 days. On day 6, acetic acid and butyric acid in the O_2-NBW groups were 37.17–53.65% and 29.48–60.54% higher than those in the DW group, respectively. On the contrary, the accumulation of intractable propionic acid makes a negative impact on methane production (Wang et al., 2023). With the addition of O_2-NBW in T1, T2 and T3 groups, the percentage of propionic acid in the total VFAs dropped from 40.68% (T0 group) to 32.70%, 35.68% and 33.64%, respectively. In brief, the provision of O_2-NBW was beneficial to the hydrolytic acidification of *E. canadensis*, which was consistent with the observations of α-glucosidase and VS as well.

pH is an intuitive parameter to assess the stability of anaerobic digestion system, in relation to VFAs. In the early phase of digestion, the system pH declined unremittingly on account of the accumulation of VFAs, and dropped to the lowest value on day 5 (Fig. S3(a)). The lowest pH values attained in disparate groups were organized sequentially as T3 > T1 > T0 > T2. Moreover, the persistent period of lower pH coincided with the period of higher VFAs. Subsequently, the pH value gradually raised and reverted to around 7.2. This might be caused by VFAs feeding methanogens for methanogenesis and the generation of ammonia nitrogen from proteolysis strengthening the buffering capacity of digestion system (Li et al., 2017).

Ammonia nitrogen concentration is connected with the decomposition of organic proteins in *E. canadensis*. From Fig. S3(b), an initial increase followed by a plateauing trend was shown for ammonia nitrogen in all digesters. On all accounts, since the concentration of ammonia nitrogen in all digesters were consistently less than 350 mg L^{-1} (<1000 mg L^{-1} in the study of Fan et al. (2021a)), no ammonia inhibition occurred.

3.4. Changes in microbial communities with O_2-NBW addition

3.4.1. Microbial community diversity

Alpha diversity, including ACE, Chao1, Shannon, Simpson, and Coverage in this study, indicates the species richness and diversity of microbial communities. As shown in Table S2, high coverage (greater than 0.994) for both bacteria and archaea were presented, illustrating high representativeness and authenticity of the sequencing results. ACE and Chao1 indices were utilized to appraise species richness, while Shannon and Simpson indices were calculated to measure species diversity. The ACE, Chao1, and Shannon indices of not only bacteria but also archaea were significantly higher in digesters supplemented with O_2-NBW, demonstrating that O_2-NBW addition might fortify the richness and diversity of bacteria and archaea in digestion system, in similarity with the observations of Wang et al. (2020a). The interpretation of this consequence could be that the microaerobic environment formed by O_2-NBs blasting facilitates the growth and activities of fast-growing facultative bacteria (Guo et al., 2015), and in addition O_2-NBW with stronger mobility of water molecules expedited the growth and metabolism of more archaea. Furthermore, the more lively microbial metabolism and complicated microbial community interactions, the greater contribution to methane production and conversion of organic components. Hence, the particular mechanisms underlying involved in variation of microbial community structure by O_2-NBW would be worthy of further investigation.

3.4.2. Microbial community structure

Changes in microbial community structure due to O_2-NBW addition

were probed, either with regard to relative abundance of bacteria (at the phylum level) or with regard to relative abundance of archaea (at the genus level). Fig. 4(a) compared the composition and relative abundance of bacteria based on phylum level under various additions of O_2-NBW or DW. An increase in the relative abundance of *Firmicutes*, *Bacteroidetes*, *Chloroflexi, Synergistetes*, *Desulfobacteria* and *Thermotogae* was observed in O_2-NBW groups compared to DW group. *Firmicutes*, which is associated with the secretion of extracellular enzymes (e.g., cellulase, protease and lipase), is known as cellulolytic bacteria as well as acetogenic bacteria in the digestion system, most of whose members are syntrophic bacteria (Garcia-Pena et al., 2011). *Firmicutes* was dominant in all groups, accounting for more than 60% of the relative abundance, however, the addition of O_2-NBW significantly magnified the relative abundance of *Firmicutes* (1.48–5.31%) in contrast with the control group. Therefore, this further confirmed that O_2-NBW provision enabled the acceleration of hydrolytic acidification of lignocellulosic biomass, as previously observed for enzyme activities and VFAs. *Bacteroidetes* is identified as proteolytic bacteria in charge of the hydrolysis of polysaccharides and proteins, while being strong oxygen-tolerant anaerobes (Vanwonterghem et al., 2014). A maximum relative abundance of *Bacteroidetes* was achieved in the T3 group (15.52%), followed by T1 (13.85%), T2 (12.79%), T0 (12.15%) groups, respectively. *Chloroflexi*, as facultative anaerobes capable of decomposing cellulose, plays a crucial part in taking full advantage of carbohydrates (Serna-Garcia et al., 2020). The presence of *Synergistetes* is supposed to be inextricably linked with amino acid degradation, with emphasis on its synergistic oxidation function that mitigates the accumulation of refractory propionate and butyrate, thereby enhancing digestive stability (Akyol et al., 2019). *Desulfobacteria* is claimed to possess great metabolic potential to accelerate the utilization and transformation of nutrients (Cheng et al., 2022). *Thermotogae* is able to fleetly make use of various monosaccharides such as glucose and polysaccharides such as xylan to maintain growth in a broad temperature range (Martinez et al., 2014). Collectively, the hydrolytic bacteria in O_2-NBW digesters were enriched, especially facultative bacteria, implying the enhancement of hydrolysis during anaerobic digestion.

In anaerobic systems, the presence and abundance of methanogenic archaea can make a direct difference to the velocity and efficiency of methane production. On the genus level, the dominant archaea found in this study were *Methanosarcina*, *Methanobacterium*, *Methanoculleus*, *Methanomassiliicoccus* and *Methanosaeta*, which were shown in Fig. 4(b). As seen, *Methanosarcina* (acetoclastic, hydrogenotrophic and mixotrophic), the most dominant archaeal genus, can flexibly utilize a variety of intermediates such as acetate, methylamine and methanol, as well as carbon dioxide and hydrogen to strengthen methane production (Ziganshina et al., 2022). Specifically, the relative abundance of *Methanosarcina* was significantly enhanced in T1, T2 and T3 groups, accounting for 85.15%, 79.10% and 81.96%, respectively, while it accounted for 75.30% in the control group (T0). In the work by Fu et al. (2016), the relative abundance of *Methanosarcina* doubled in optimal microaerobic conditions. Hence, it was deduced that the microaerobic environment constructed by O_2-NBW supplementation could make a prominent contribution to the enrichment of oxytolerant methanogens, leading to an increase in methane. Moreover, hydrogenotrophic *Methanobacterium* possesses the ability to expend hydrogen and carbon dioxide to generate methane, especially the performance of tolerance to restricted oxygen (Zitomer and Shrout, 1998). Apparently, the relative abundance of *Methanobacterium* in T1, T2 and T3 digesters with O_2-NBW were 4.55%, 5.93%, and 5.55%, respectively, higher than that in T0 (3.61%) digesters with DW. Similarly, it was reported that the relative abundance of *Methanobacterium* was as high as 10.1% after adding O_2-NBW, compared with only 1.2% in the control group (Wang et al., 2020b). As mentioned above, these findings further represented that the supply of O_2-NBW showed positive benefits on the methanogenic process. *Methanoculleus*, a hydrogenotrophic methanogen reported to augment in quantitative terms due to microaeration (Tang et al., 2004), was observed with the rise in relative abundance of 5.66–5.95% in the O_2-NBW supplemented groups versus 4.40% in the control group. Hydrogenotrophic *Methanomassiliicoccus* is positively correlated with biogas production during anaerobic digestion and can establish a symbiotic relationship with sulfate-reducing bacteria producing hydrogen (Zhang et al., 2022), with the relative abundances as 2.55%, 3.04%, 3.71%, and 4.98% in T0, T1, T2, and T3, respectively. *Methanosaeta* is a rigorous aceticlastic methanogen utilizing acetate as its sole energy source and own a higher affinity for acetate than *Methanosarcina* (Chen and He, 2015). Concretely, the relative abundance of *Methanosaeta* was elevated in O_2-NBW digesters, possibly due to the more acetate produced by microorganisms prior to methanogenesis. Taken together, the archaeal community structure could be appropriately altered to achieve excellent performance of anaerobic digestion, with O_2-NBW supplementation, especially the enrichment of oxytolerant methanogens to guarantee admirable methanogenesis.

3.4.3. Potential functions of microbial communities

According to the KEGG pathway database, Phylogenetic Investigation of Communities by Reconstruction of Unobserved States (PICRUSt) was applied to predict the primary functions of bacteria and archaea. Seen from Fig. 5(a), the majority of predicted sequences in bacterial domain were clustered into four chief categories, including metabolism (76.24–76.86%), genetic information processing (7.93–8.07%), environmental information processing (5.83–6.11%) and cellular processes (4.73–4.82%). In metabolism cluster, the most abundant functional category was carbohydrate metabolism (T0: 10.35%; T1: 10.52%; T2: 10.37%; T3: 10.46%), followed by amino acid metabolism (T0: 7.25%; T1: 7.31%; T2: 7.27%; T3: 7.38%) as well as energy metabolism (T0: 4.30%; T1: 4.43%; T2: 4.36%; T3: 4.41%) for bacteria, which were

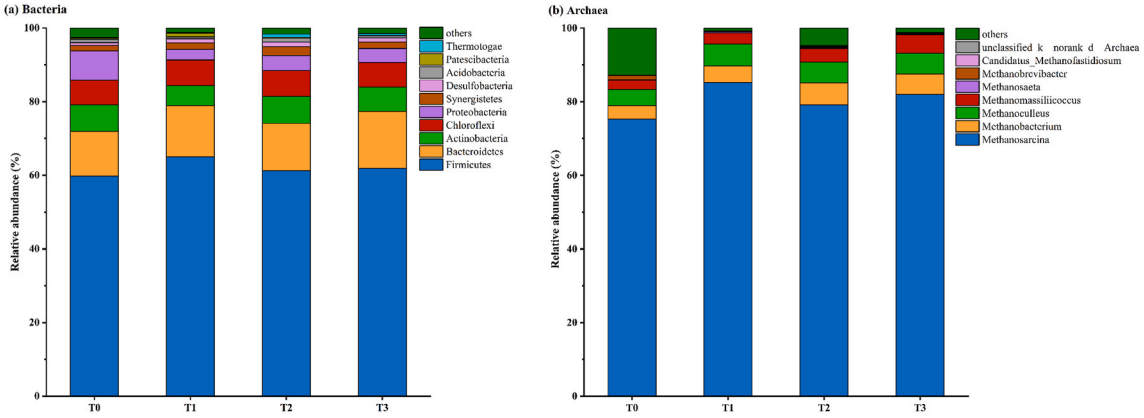


Fig. 4. Relative abundances of diverse bacterial communities at phylum level (a) and archaeal communities at genus level (b) on day 30 of anaerobic digestion.

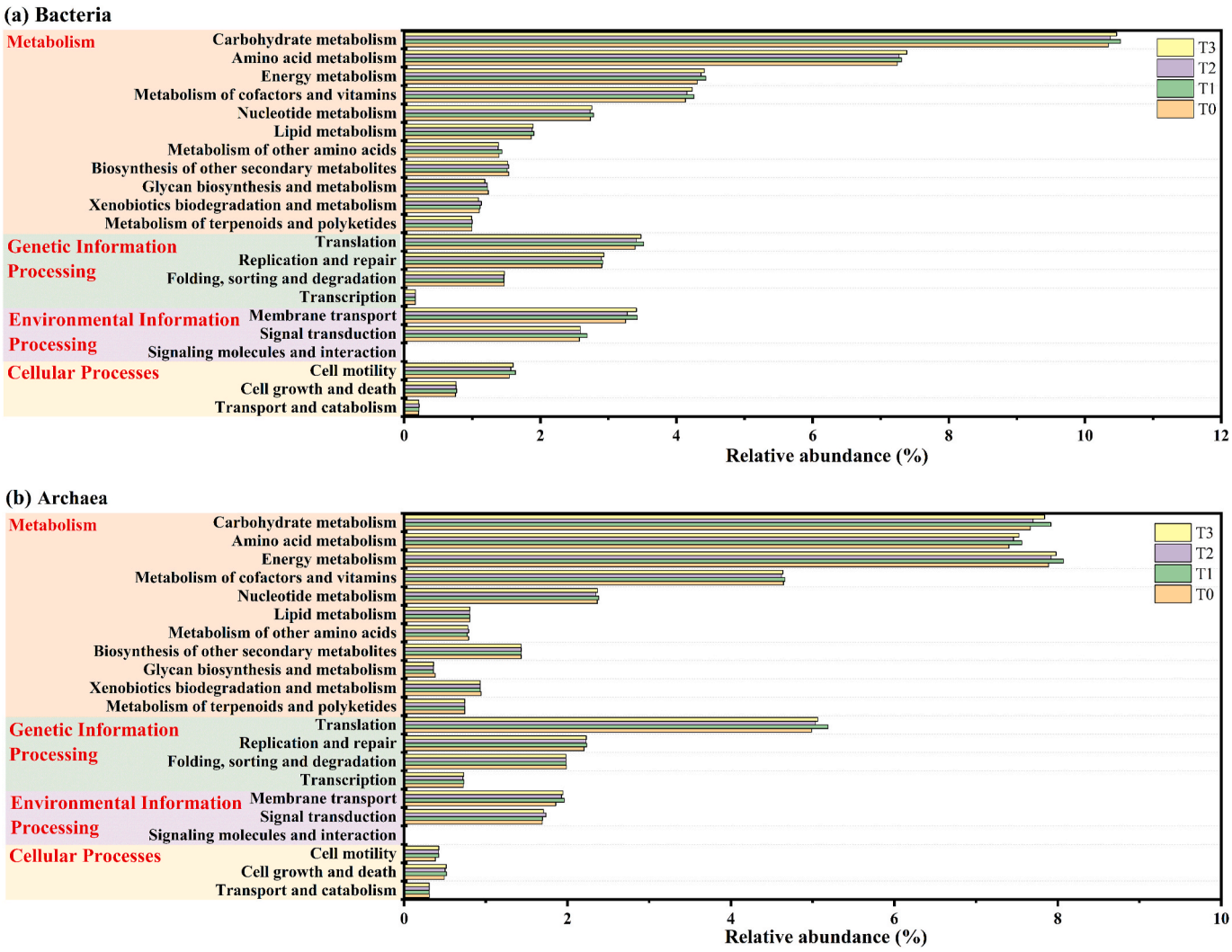


Fig. 5. (a) Bacterial and (b) Archaeal KEGG metabolic pathway.

connected with methanogenic performance during anaerobic digestion (Guo et al., 2015). Carbohydrate metabolism can make an infinite contribution to the hydrolysis of lignocellulose to provide a variety of substrates (Toledo et al., 2017). Amino acid metabolism furnishes bacteria with essential carbon sources and energy, while energy metabolism contributes to bacterial growth and reproduction (Li et al., 2022). Enhanced metabolic activity in the O_2-NBW groups might be due to the addition of O_2-NBW stimulating microorganisms and expediting the metabolism of organic matters in anaerobic digestion system. Furthermore, translation (T0: 3.39%; T1: 3.52%; T2: 3.41%; T3: 3.48%) was the principal pathway of genetic information processing for bacteria, while membrane transport (T0: 3.25%; T1: 3.43%; T2: 3.28%; T3: 3.41%) and cellular motility (T0: 1.54%; T1: 1.63%; T2: 1.57%; T3: 1.60%) were the dominating approaches for environmental information processing and cellular processes, respectively. Results indicated that the addition of O_2-NBW could strengthen bacterial protein biosynthesis, membrane transport, and cell motility, thereby acquiring better functionality of the anaerobic digestion system.

Predicted functions of archaea were identified into four main pathways as well (Fig. 5(b)). In archaeal metabolism (79.98–80.47%) cluster, the performance of carbohydrate metabolism, amino acid metabolism and energy metabolism in the O_2-NBW groups was superior to that of the control group. Among genetic information processing (7.93–8.07%), the functional variety with the highest relative abundance was for translation that could convert sequence bases of mRNA into amino acids in polypeptide chains or proteins, allowing for well expression of heredity information within cells (Ye et al., 2018). The relative abundance of archaeal translation in digesters with O_2-NBW was 0.04–0.19% higher than that in digesters with DW. Environmental information processing (3.45–3.66%) of archaea principally involved membrane transport, which was the foundation of organismal information exchange and energy transformation (Li et al., 2022), with 1.86%, 1.96%, 1.93% and 1.94% in T0, T1, T2 and T3 groups, respectively. In addition, compared to T0, the relative abundance of archaea cellular processes in T1, T2 and T3 groups were increased by 0.03%, 0.01% and 0.08%, respectively. With O_2-NBW supplementation, the enhancement of archaeal metabolic activities during anaerobic digestion was achieved, thus making the methanogenic performance of digestion system greater.

3.5. Carbon balance analysis

The system boundary for carbon balance analysis was an integral anaerobic digestion system of *E. canadensis*. Total carbon input and output of the digestion system were monitored by utilizing elemental analysis and based on the stoichiometric content of carbon in gas (Sosnowski et al., 2008). Thereinto, the input incorporated the carbon distributed in *E. canadensis* and inoculum sludge, while the output consisted of the carbon contained in CH_4, CO_2, biogas residue, and others such as microbial carbon consumption. All carbon distributions

were shown as a percentage of total carbon input to the digestion system (Qi et al., 2022). As shown in Fig. 6, there were discrepancies in carbon flow among the four groups under disparate additions of O_2-NBW or DW. An increase of 4.30–16.93% in the conversion of carbon into methane was observed in O_2-NBW groups in contrast with DW group, which illustrated that supplementation of O_2-NBW was beneficial to the efficient flow of organic carbon in *E. canadensis* to methane. Particularly, the most efficient carbon utilization of 40.94% (CH_4 and CO_2) during anaerobic digestion was achieved in the T1 group with one-time addition of O_2-NBW on day 0. However, a semblable trend of carbon flow was presented in each group, with the distribution to biogas residue (49.19–53.23%) > CH_4 (22.09–25.83%) > CO_2 (12.98–15.11%) > others (9.87–11.70%). Collectively, the addition of O_2-NBW in anaerobic digestion system could accomplish effective methanation utilization of carbon in biomass.

3.6. Preliminary analysis of interrelated mechanisms

In accordance with research results, O_2-NBW supplementation could achieve an increase in methane production. O_2-NBW with O_2-NBs has higher zeta potential, residence time, internal pressure and gas solubility than DW (Fan et al., 2020a, 2020b). Due to the disturbance of hydrogen bonding network of water via O_2-NBs, the permeability of substrates into internal area of biofilms was enhanced, and hence the efficiency of mass transfer of nutrients and microbial activity during anaerobic digestion would be heightened by means of O_2-NBW with strong mobility of water molecules (Wang et al., 2019). Bursting of O_2-NBs as well as release of oxygen, interestingly, might create a microaerobic environment to reinforce the proliferation of facultative bacteria and activity of corresponding enzymes (Wang et al., 2020b), thereby facilitating the conversion of substrates in hydrolytic acidification process, especially accelerating the manufacture or consumption of intermediates such as VFAs. Additionally, there is a symbiotic synergy among diverse microbial groups during anaerobic digestion under microaerobic conditions, resulting in stronger stability (Fu et al., 2016). In particular, microorganisms in digestion system generally tend to form flocs in which strict anaerobes are surrounded by facultative bacteria (Liu et al., 2009). Facultative bacteria in outer layer can not only obliterate reactive oxygen species (ROS), but also deplete oxygen to decompose complicated organic substances into disparate intermediates, which are then utilized as substrates by methanogens (Stewart and Franklin, 2008). As a consequence, the amount of remanent oxygen to the interior drops to the level within the tolerance limits of obligate anaerobic methanogens (Nguyen and Khanal, 2018). Therefore, facultative bacteria are regard as both physical and biological barriers against oxygen in the outer layer, enabling the survival of strict anaerobic archaea and bacteria to perform their duties efficiently in a microaerobic environment (Botheju and Bakke, 2011). On the other hand, under oxidative stress conditions, a few strict anaerobes, such as *Methanobrevibacter*, *Methanobacterium*, and *Methanosarcina*, could manufacture antioxidative enzymes to acclimatize themselves to the oxidative environment, thereby enhancing their tolerance to oxygen (Brioukhanov and Netrusov, 2007; Horne and Lessner, 2013). Furthermore, O_2-NBW could advance the ETS activity through the release of oxygen, as well as drive ATP synthesis in virtue of the TCA cycle (Wang et al., 2021). Last but not least, the addition of O_2-NBW was beneficial to the enhancement of methanogen activity in the methanogenic phase to further boost methane production.

4. Conclusions

This study investigated the methanogenic performance of *E. canadensis* in anaerobic digestion system via diverse O_2-NBW addition times. Supplementation of O_2-NBW enhanced methane production by 10.70–16.17%, while achieving the maximum cumulative methane production of 343.18 mL g^{-1} VS in the digester with one-time addition

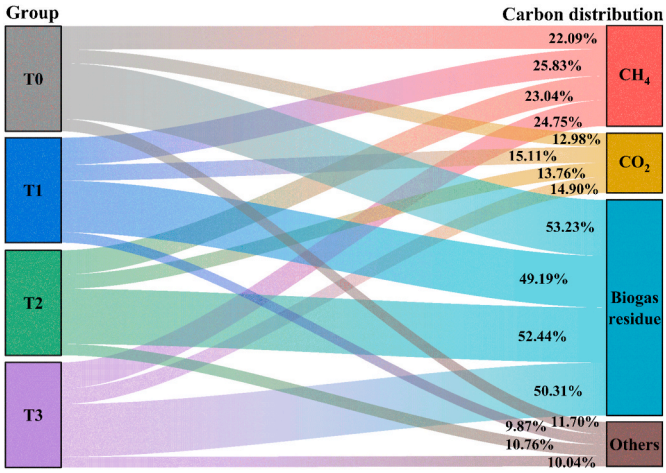


Fig. 6. Carbon flow for various additions of O_2-NBW and DW.

of O_2-NBW on day 0. An increase in enzyme activities was observed in the O_2-NBW digesters as well, about 2–90%, especially the activity of coenzyme F_{420} was significantly increased by 14.69–89.95% in comparison to the DW digester. Moreover, the addition of O_2-NBW could advance the removal rate of VS as well as the generation and consumption of VFAs in hydrolytic acidification process. An enrichment of not only hydrolytic bacteria (facultative bacteria) but also methanogenic archaea (oxytolerant methanogens) was presented, which might be responsible for the accelerated hydrolysis and methanogenesis resulting from the addition of O_2-NBW. According to KEGG analysis of bacterial and archaeal communities, the predominant functions augmented by O_2-NBW supplementation were associated with carbohydrate metabolism, amino acid metabolism, energy metabolism, translation, membrane transport, and cell motility. Last but not least, the provision of O_2-NBW in the anaerobic digestion system achieved efficient utilization of carbon in biomass based on carbon balance analysis.

CRediT authorship contribution statement

Legeng Pei: Conceptualization, Formal analysis, Methodology, Visualization, Writing – original draft. **Yingjin Song:** Funding acquisition, Writing – review & editing. **Guanyi Chen:** Resources, Supervision, Validation. **Lan Mu:** Writing – review & editing. **Beibei Yan:** Project administration. **Teng Zhou:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.chemosphere.2024.141732.

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